

EINDHOVEN AP, Netherlands

WMO: 063700

Lat: 51.4497N

Lon: 5.3769E

Elev: 23

StdP: 101.05

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-4.7	-10.5	1.5	-4.7	-8.0	1.9	-3.5	12.4	9.5	11.3	9.4	2.6	70	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.1	20.0	27.9	19.2	25.9	18.3	21.1	27.6	20.2	25.9	19.2	24.2	3.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	23.4	18.1	13.1	22.3	17.3	12.4	21.3	61.3	27.9	57.9	26.0	54.8	24.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.8	7.7	-10.4	33.9	3.4	1.8	-12.8	35.2	-14.8	36.2	-16.7	37.2	-19.2	38.4
			-10.7	23.2	3.4	1.0	-13.1	23.9	-15.1	24.4	-17.0	25.0	-19.5	25.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	3.3	3.9	6.5	10.0	13.7	16.5	18.6	18.2	15.0	11.3	6.9	3.9		
	DBStd	6.55	4.31	4.03	3.56	3.80	3.66	3.30	3.33	2.98	2.84	3.44	3.58	4.21		
	HDD10.0	871	209	171	116	46	8	0	0	0	1	27	103	191		
	HDD18.3	2902	466	403	366	250	152	74	37	39	107	219	342	447		
	CDD10.0	1124	1	2	9	47	121	195	268	253	150	66	11	3		
	CDD18.3	114	0	0	0	1	7	19	47	34	7	1	0	0		
	CDH23.3	1114	0	0	0	18	81	208	435	313	56	3	0	0		
	CDH26.7	321	0	0	0	1	13	57	139	101	10	0	0	0		
(14) Wind	WSAvg	3.6	4.3	4.2	4.0	3.6	3.5	3.4	3.2	3.0	3.2	3.6	3.7	4.2		
(15) Precipitation	PrecAvg	812	70	67	52	45	61	72	79	79	65	65	75	82		
	PrecMax	1037	145	147	103	85	113	207	144	169	167	178	130	156		
	PrecMin	616	3	14	4	0	19	13	26	23	10	22	14	27		
	PrecStd	95	34	33	28	21	26	38	35	45	41	29	29	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	15.0	19.6	25.9	28.5	31.7	33.6	32.4	28.0	23.1	17.1	13.7		
		MCWB	11.2	10.0	12.6	15.0	18.4	20.8	20.7	21.1	19.2	17.3	14.0	11.9		
	2%	DB	11.7	12.3	16.5	22.4	25.5	28.1	30.3	29.5	24.8	19.9	14.8	12.5		
		MCWB	10.1	9.5	10.9	13.8	16.6	19.1	20.2	19.9	18.1	15.9	12.7	11.0		
	5%	DB	10.3	11.0	14.1	19.5	23.3	25.6	27.8	26.9	22.3	17.8	13.5	11.2		
		MCWB	8.9	8.9	10.1	12.7	15.6	18.0	19.4	19.3	17.2	15.0	11.8	9.9		
	10%	DB	9.0	9.7	12.2	16.9	20.9	23.4	25.6	24.3	20.2	16.3	12.1	9.8		
		MCWB	7.7	8.1	9.3	11.5	14.7	17.0	18.6	18.1	16.0	14.0	10.7	8.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.2	13.2	15.9	19.7	22.0	22.6	22.5	20.3	17.8	14.4	12.5		
		MCDB	12.6	13.0	17.4	22.4	25.9	28.8	29.4	29.5	25.8	21.6	16.0	13.2		
	2%	WB	10.3	10.1	11.8	14.4	17.9	20.1	21.2	20.9	18.9	16.5	13.1	11.3		
		MCDB	11.4	11.7	15.0	20.4	23.5	26.0	27.8	27.3	23.3	19.1	14.6	12.3		
	5%	WB	9.0	9.3	10.7	13.3	16.4	18.9	20.3	19.9	17.8	15.4	12.1	10.1		
		MCDB	10.2	10.8	13.3	18.6	21.2	24.0	26.1	25.2	21.2	17.5	13.3	11.1		
	10%	WB	7.9	8.1	9.6	12.1	15.2	17.7	19.2	18.8	16.7	14.4	10.8	8.8		
		MCDB	8.9	9.5	11.9	16.1	19.7	22.4	24.2	23.2	19.5	16.1	11.9	9.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.2	8.1	10.0	10.1	10.2	9.9	10.1	9.2	7.7	5.8	4.9		
		MCDBR	5.5	7.6	11.7	14.1	13.9	14.0	13.6	13.9	12.0	9.3	6.5	5.3		
	5% WB	MCWBR	4.8	5.4	7.0	7.0	6.6	6.5	5.7	6.0	6.2	5.9	4.9	4.6		
		MCWBR	5.4	6.4	9.7	12.1	11.7	12.1	12.0	12.2	10.5	8.6	6.0	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.360	0.393	0.423	0.431	0.427	0.435	0.425	0.398	0.373	0.349	0.333		
		taud	2.344	2.307	2.221	2.177	2.203	2.229	2.211	2.260	2.325	2.372	2.352	2.319		
	Ebn at Noon		663	750	791	812	825	831	815	802	781	726	639	596		
		Edh at Noon	65	87	113	133	136	133	134	121	102	81	64	57		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.43	2.59	4.17	5.00	5.41	5.07	4.33	3.20	1.88	0.88	0.57		
	RadStd		0.09	0.28	0.37	0.45	0.42	0.43	0.61	0.39	0.29	0.15	0.09	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (87 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page